

REGRESSION CAPSTONE PROJECT REPORT

PROJECT TITLE: Life Expectancy Prediction in India using Linear Regression

1. OBJECTIVE:

- Analyze life expectancy trends in India over the years
- Study the effect of infant mortality and adult mortality on life expectancy
- Apply linear regression to model and interpret these relationships

2. DATASET DESCRIPTION:

- Source: life.csv containing global health and demographic data
- Key attributes: Year, Life Expectancy, Infant Mortality, Adult Mortality, and other health indicators
- Filtered: Records specific to India
- Focus Variables: Infant Mortality, Adult Mortality, and Life Expectancy

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 2938 entries, 0 to 2937
Data columns (total 22 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Country                               2938 non-null   object
1   Year                                  2938 non-null   int64
2   Status                                2938 non-null   object
3   Life expectancy                       2928 non-null   float64
4   Adult Mortality                       2928 non-null   float64
5   infant deaths                         2938 non-null   int64
6   Alcohol                               2744 non-null   float64
7   percentage expenditure                2938 non-null   float64
8   Hepatitis B                           2385 non-null   float64
9   Measles                               2938 non-null   int64
10  BMI                                    2904 non-null   float64
11  under-five deaths                     2938 non-null   int64
12  Polio                                 2919 non-null   float64
13  Total expenditure                     2712 non-null   float64
14  Diphtheria                           2919 non-null   float64
15  HIV/AIDS                              2938 non-null   float64
16  GDP                                    2490 non-null   float64
17  Population                            2286 non-null   float64
18  thinness 1-19 years                   2904 non-null   float64
19  thinness 5-9 years                    2904 non-null   float64
20  Income composition of resources       2771 non-null   float64
21  Schooling                             2775 non-null   float64
dtypes: float64(16), int64(4), object(2)
memory usage: 505.1+ KB
None
```

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3. METHODOLOGY USED:

1. Data Preprocessing:

- Filtered dataset for India
- Handled missing values and ensured data consistency

]:

	Year	infant deaths	Adult Mortality	Alcohol
1201	2000.0	1800.0	224.0	0.93
1200	2001.0	1800.0	222.0	1.00
1199	2002.0	1700.0	219.0	1.10
1198	2003.0	1700.0	216.0	1.19
1197	2004.0	1600.0	214.0	1.20
1196	2005.0	1500.0	211.0	1.27
1195	2006.0	1500.0	28.0	1.37
1194	2007.0	1400.0	26.0	1.59
1193	2008.0	1300.0	23.0	1.93
1192	2009.0	1300.0	2.0	2.50
1191	2010.0	1200.0	196.0	2.77
1190	2011.0	1100.0	193.0	3.00
1189	2012.0	1100.0	19.0	3.10
1188	2013.0	1000.0	187.0	3.11
1187	2014.0	957.0	184.0	3.07

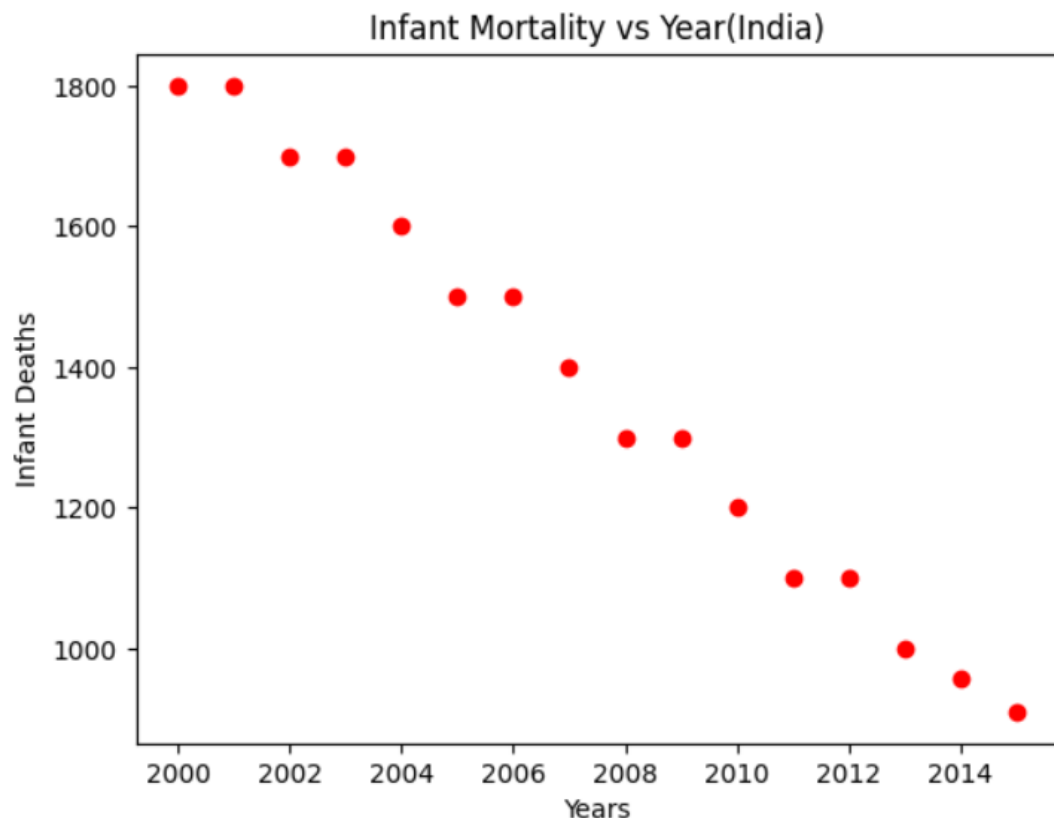
:

	Year	infant deaths	Adult Mortality	Alcohol
count	16.000000	16.000000	16.000000	16.000000
mean	2007.500000	1366.687500	146.562500	2.108304
std	4.760952	303.160402	89.558151	1.076132
min	2000.000000	910.000000	2.000000	0.930000
25%	2003.750000	1100.000000	27.500000	1.197500
50%	2007.500000	1350.000000	190.000000	1.760000
75%	2011.250000	1625.000000	214.500000	3.017500
max	2015.000000	1800.000000	224.000000	4.602861

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2. Exploratory Data Analysis:

- Visualized mortality and life expectancy trends over years
- Identified strong correlations



3. Modeling:

- Applied Linear Regression using scikit-learn
- Evaluated relationship between predictor (years) and targets (infant mortality & adult mortality)

```
#training model
model= LinearRegression().fit(x_year, y_infant_deaths)
model
```

```
LinearRegression ⓘ ?
  Parameters
```

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4. SOURCE CODE:

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.impute import SimpleImputer

#importing dataset
df=pd.read_csv('Life.csv')
print(df.info())

# Select numeric features
numeric_features = df.select_dtypes(include=np.number)

# Impute missing values in numeric features
imputer = SimpleImputer(strategy='mean')
imputed_data = pd.DataFrame(
    imputer.fit_transform(numeric_features),
    columns=numeric_features.columns
)

# Combine: imputed numeric data + Country column
df = pd.concat([imputed_data, df[['Country']], axis=1)

# Check remaining null values
df.isnull().sum()

#data cleaning
india= df[df['Country'].str.strip()=='India'].sort_values('Year')
india[['Year','infant deaths','Adult Mortality','Alcohol']].head(15)

#data analysis
india[['Year','infant deaths','Adult Mortality','Alcohol']].describe()

#Visual representation and analysis(for Year and infant mortality)
x_year= india['Year'].values
y_infant_deaths= india['infant deaths'].values
plt.scatter(x_year, y_infant_deaths, color= 'red')
plt.xlabel('Years')
```

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```
plt.ylabel('Infant Deaths')
plt.title('Infant Mortality vs Year(India)')
plt.show()

#data exploration
x_year = x_year.reshape(-1,1)
x_year

#training model
model= LinearRegression().fit(x_year, y_infant_deaths)
model

#model predictions
prediction= model.predict(x_year)
prediction

#Data representation
plt.scatter(x_year,y_infant_deaths, color='red')
plt.plot(x_year,prediction, color='green')
plt.xlabel('Years')
plt.ylabel('Infant Deaths')
plt.title('Model Predictions without train test split(India)')
plt.show()

#splitting data into training and testing(for year and infant mortality)
x_train, x_test, y_train, y_test = train_test_split(
    x_year, y_infant_deaths, train_size=0.7, test_size=0.3, random_state=42
)
print(f"X_train shape ={x_train.shape}")
print(f"Y_train shape ={y_train.shape}")
print(f"X_test shape ={x_test.shape}")
print(f"Y_test shape ={y_test.shape}")

# data representation for training data(for year and infant mortality)
plt.scatter(x_train,y_train, color='green')
plt.xlabel('Years')
plt.ylabel('Infant Deaths')
plt.title('Training data')
plt.show()

# data representation for testing data(for year and infant mortality)
plt.scatter(x_test,y_test, color='red')
plt.xlabel('Years')
plt.ylabel('Infant Deaths')
```

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```
plt.title('Testing data')
plt.show()

#training the model(for infant deaths and year)
model=LinearRegression()
model.fit(x_train,y_train)

#model predictions
y_pred=model.predict(x_test)
y_pred

# 1. Weight (Slope)
print("Weight (Slope) for Infant Mortality Prediction:", model.coef_)

# 2. Bias (Intercept)
print("Bias (Intercept) for Infant Mortality Prediction:", model.intercept_)

from sklearn.metrics import mean_absolute_error,mean_squared_error
from sklearn.metrics import r2_score
# 1. Mean Squared Error (MSE)
mse_inf = mean_squared_error(y_test, y_pred)
print("Mean Squared Error (MSE) for Infant Mortality Prediction:", mse_inf)

# 2. Mean Absolute Error (MAE)
mae_inf = mean_absolute_error(y_test, y_pred)
print("Mean Absolute Error (MAE) for Infant Mortality Prediction:", mae_inf)

# 2. R-square Error
r2_inf= r2_score(y_test, y_pred)
print("R2 Score for Infant Mortality Prediction:", r2_inf)

#Visual representation for Infant Mortality prediction
plt.scatter(x_test,y_test, color= 'red')
plt.plot(x_test, model.predict(x_test), color='blue')
plt.title('Infant Mortality vs Years (India)')
plt.xlabel('Years')
plt.ylabel('Infant Deaths')
plt.show()

#Visual representation and analysis(for Year and adult mortality)
x_year= india['Year'].values
```

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```
y_adult_mortality= india['Adult Mortality'].values
plt.scatter(x_year, y_adult_mortality, color= 'red')
plt.xlabel('Years')
plt.ylabel('Adult Mortality')
plt.title('Adult Mortality vs Year(India)')
plt.show()
```

```
#data exploration
x_year = x_year.reshape(-1,1)
x_year
```

```
#splitting data into training and testing(for year and adult mortality)
x_train, x_test, y_train, y_test = train_test_split(
    x_year, y_adult_mortality, train_size=0.7, test_size=0.3, random_state=3
)
print(f"X_train shape ={x_train.shape}")
print(f"Y_train shape ={y_train.shape}")
print(f"X_test shape ={x_test.shape}")
print(f"Y_test shape ={y_test.shape}")
```

```
# data representation for training data(for year and adult mortality)
plt.scatter(x_train,y_train, color='green')
plt.xlabel('Years')
plt.ylabel('Adult Mortality')
plt.title('Training data')
plt.show()
```

```
# data representation for testing data(for year and adult mortality)
plt.scatter(x_test,y_test, color='red')
plt.xlabel('Years')
plt.ylabel('Adult Mortality')
plt.title('Testing data')
plt.show()
```

```
#training the model(for adult mortality and year)
model_adu=LinearRegression()
model_adu.fit(x_train,y_train)
```

```
#model predictions
y_pred=model_adu.predict(x_test)
```

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```
y_pred

# 1. Weight (Slope)
print("Weight (Slope) for Adult Mortality Prediction:", model_adu.coef_)

# 2. Bias (Intercept)
print("Bias (Intercept) for Adult Mortality Prediction:", model_adu.intercept_)

from sklearn.metrics import mean_absolute_error, mean_squared_error
from sklearn.metrics import r2_score
# 1. Mean Squared Error (MSE) for Adult Mortality
mse_adu = mean_squared_error(y_test, y_pred)
print("Mean Squared Error (MSE) for Adult Mortality Prediction:", mse_adu)

# 2. Mean Absolute Error (MAE) for Adult Mortality
mae_adu = mean_absolute_error(y_test, y_pred)
print("Mean Absolute Error (MAE) for Adult Mortality Prediction:", mae_adu)

# 2. R-square Error for Adult Mortality
r2_adu = r2_score(y_test, y_pred)
print("R2 Score for Adult Mortality Prediction:", r2_adu)

# Visual representation for Adult Mortality prediction
plt.scatter(x_test, y_test, color='red')
plt.plot(x_test, model_adu.predict(x_test), color='blue')
plt.title('Adult Mortality vs Years (India)')
plt.xlabel('Years')
plt.ylabel('Adult Mortality')
plt.show()

# data cleaning
rus = df[df['Country'].str.strip() != 'Russian Federation'].sort_values('Year')
rus[['Year', 'infant deaths', 'Adult Mortality']].head(15)

# data analysis
rus[['Year', 'infant deaths', 'Adult Mortality']].describe()

# Visual representation and analysis (for Year and infant mortality)
x_year = rus['Year'].values
```

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```
y_infant_deaths= rus['infant deaths'].values
plt.scatter(x_year, y_infant_deaths, color= 'red')
plt.xlabel('Years')
plt.ylabel('Infant Deaths')
plt.title('Infant Mortality vs Year(Russia)')
plt.show()
```

```
y_infant_deaths
```

```
#data exploration
x_year = x_year.reshape(-1,1)
x_year
```

```
model_rus=LinearRegression().fit(x_year,y_infant_deaths)
prediction=model_rus.predict(x_year)
```

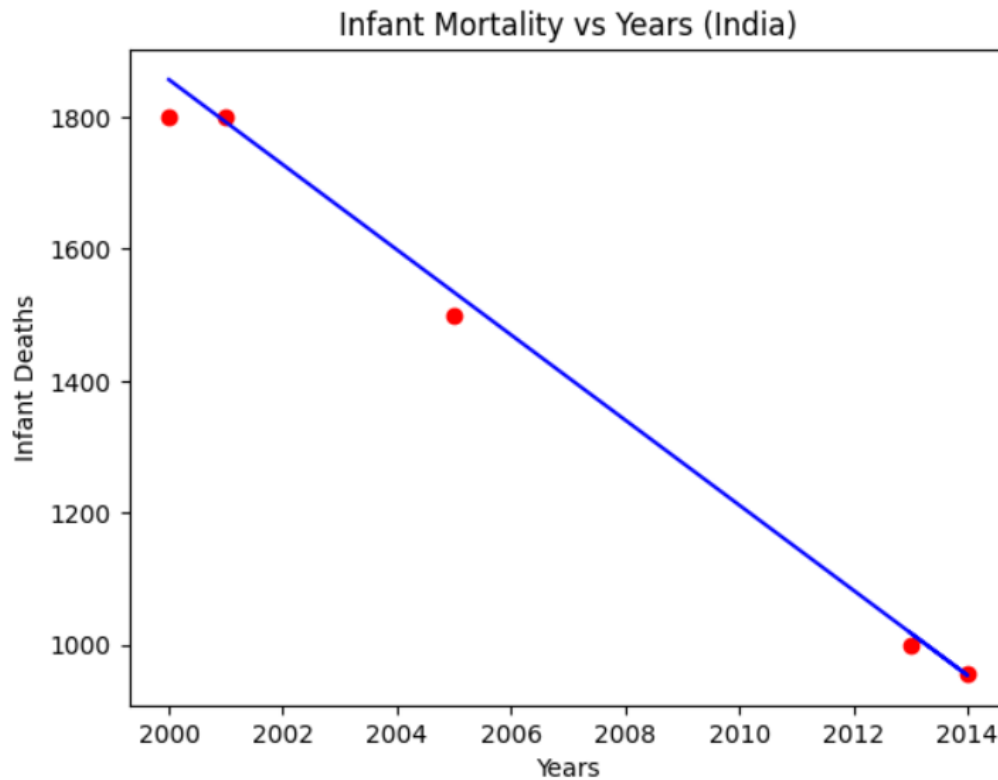
```
plt.scatter(x_year,y_infant_deaths,color='red')
plt.plot(x_year,prediction,color='green')
plt.xlabel('Years')
plt.ylabel('Infant Deaths')
plt.title('Model Predictions without train test split(Russia)')
plt.show()
```

```
from sklearn.metrics import mean_absolute_error,mean_squared_error
from sklearn.metrics import r2_score
# 1. Mean Squared Error (MSE) for Adult Mortality
mse_rus = mean_squared_error(y_infant_deaths, prediction)
print("Mean Squared Error (MSE) for Infant Mortality Prediction:", mse_rus)
```

```
# 2. Mean Absolute Error (MAE) for Adult Mortality
mae_rus = mean_absolute_error(y_infant_deaths, prediction)
print("Mean Absolute Error (MAE) for Infant Mortality Prediction:", mae_rus)
```

```
# 2. R-square Error for Adult Mortality
r2_rus= r2_score(y_infant_deaths, prediction)
print("R2 Score for Infant Mortality Prediction:", r2_rus)
```

5. RESULT DISCUSSION:

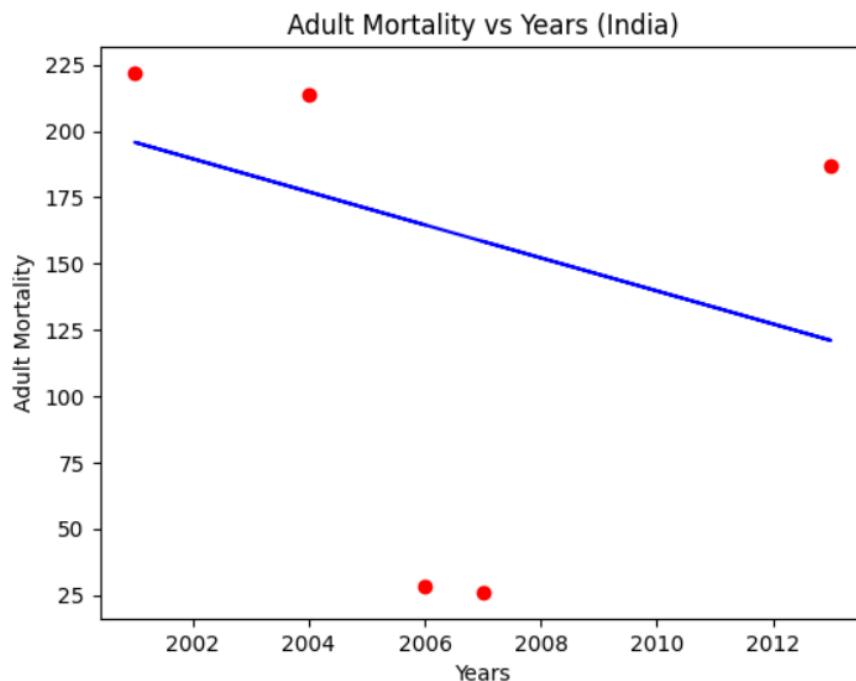


Mean Squared Error (MSE) for Infant Mortality Prediction: 954.9577005333565

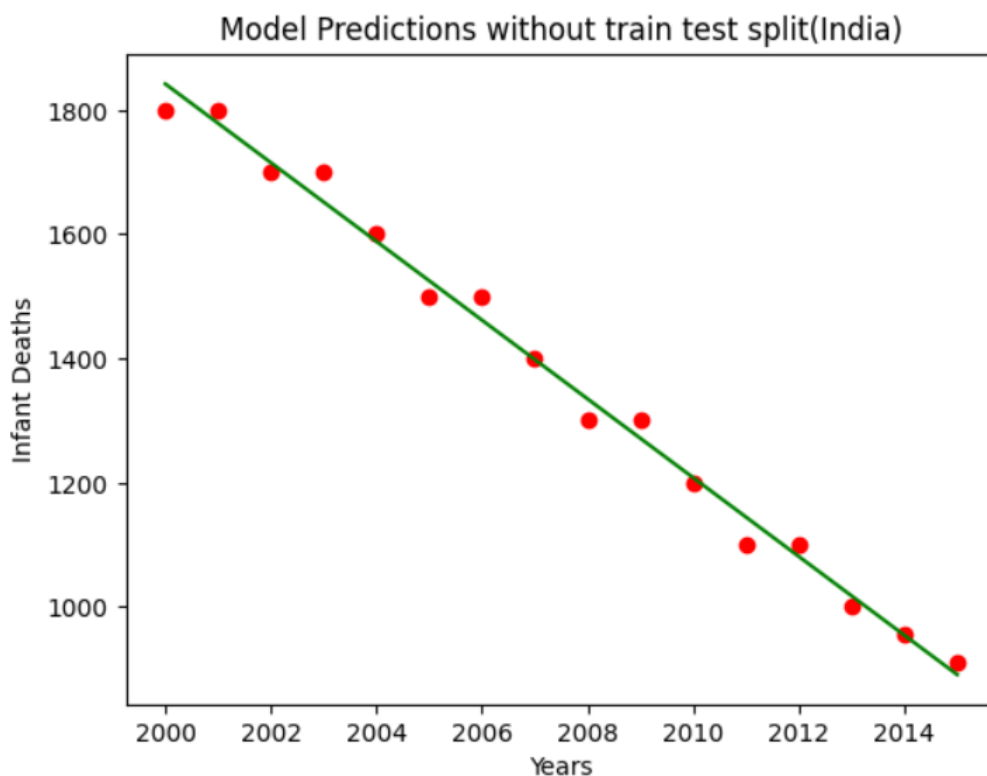
Mean Absolute Error (MAE) for Infant Mortality Prediction: 24.0305084745778

R^2 Score for Infant Mortality Prediction: 0.9930355979081266

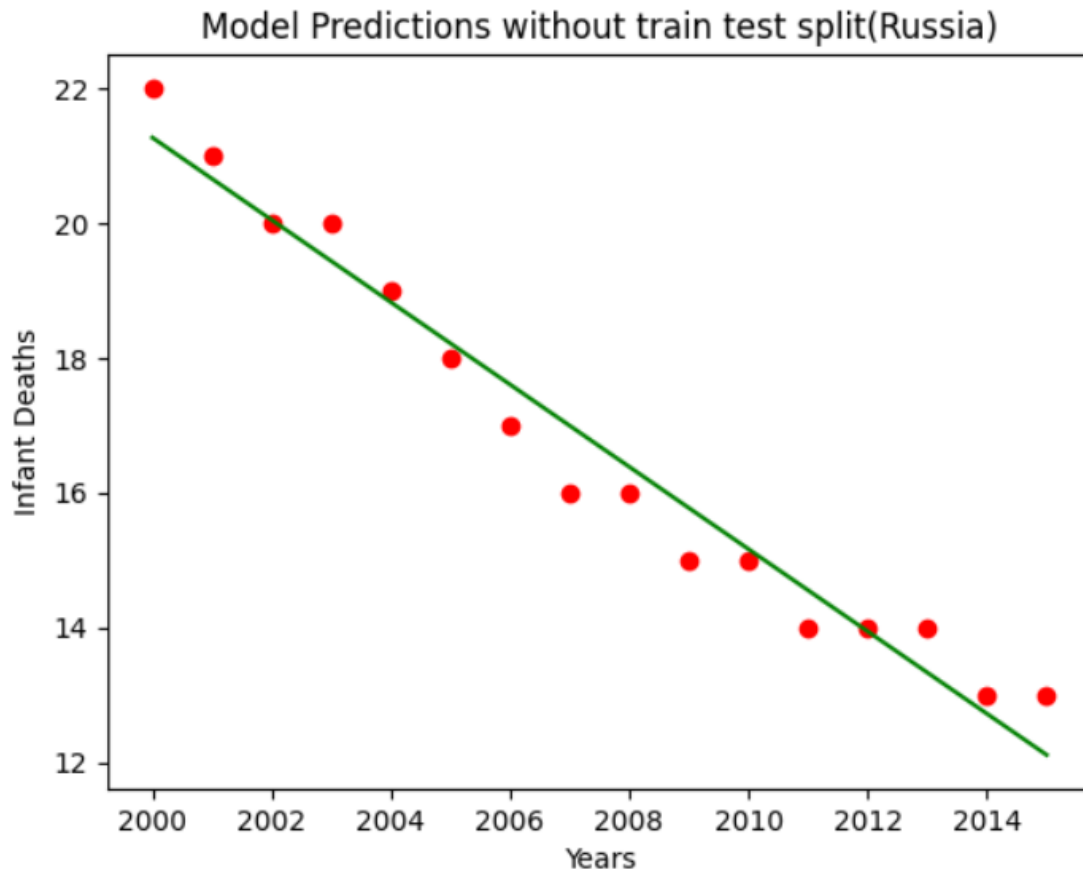
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Mean Squared Error (MSE) for Adult Mortality Prediction: 8521.930387143855
Mean Absolute Error (MAE) for Adult Mortality Prediction: 79.62162162162167
 R^2 Score for Adult Mortality Prediction: -0.06943347762527297



Applied Machine Learning Lab- MCA 2025 – Semester 1



Mean Squared Error (MSE) for Infant Mortality Prediction: 0.30009191176473327
Mean Absolute Error (MAE) for Infant Mortality Prediction: 0.46507352941178226
 R^2 Score for Infant Mortality Prediction: 0.963469553299205

a. Infant Mortality (India & Russia)

- i. Clear **downward trend** over the years (2000–2015).
- ii. Regression line fits the data very well.
- iii. Error metrics:
 1. **Low MSE & MAE**
 2. **High R^2 (≈ 0.99)** → Excellent predictive performance.
- iv. Model is highly reliable for infant mortality prediction.

b. Adult Mortality (India)

- i. Data points are **scattered and inconsistent** (sudden drops and rises).
- ii. Regression line does not fit well.
- iii. Error metrics:
 1. **High MSE & MAE**
 2. **Negative R^2** → Model performs worse than a simple mean prediction.
- iv. Model is unreliable for adult mortality prediction.

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c. Comparison (Infant vs. Adult Mortality)

- i. Infant mortality models are **highly accurate** and show steady improvement over time.
- ii. Adult mortality models are **poorly accurate** due to anomalies and non-linear patterns in the data.

6. CONCLUSION:

The improvement in life expectancy is largely driven by the significant decrease in infant and adult mortality rates. Reduced infant mortality reflects better child healthcare, vaccinations, and nutrition, while lower adult mortality indicates progress in managing diseases and healthcare access. Together, these reductions result in longer average lifespans and reflect overall advancements in public health and living conditions.

Visual Summary:

- The upward trend in life expectancy is evident.
- Infant and adult mortality show a clear negative relationship with life expectancy.
- These visuals confirm how health improvements directly enhanced longevity in India.

